

Problem 5

[15p] Heaps over binary tree and over arrays

For the next two items you can either represent the binary tree using the mathematical representation (i.e., $BT = Nil \mid \langle root, BT, BT \rangle$) or the in memory representation (i.e., $BT = \langle root: Node \rangle$, $Node: \langle data, left: BT, right: BT \rangle$).

1. [5p] Write an algorithm that converts a binary tree into an array. What are the preconditions over the binary tree such that the resulting array has the max-heap property?

```
1  private int countNodes(Node curr) {
2      if (curr == null) return 0;
3      return 1 + countNodes(curr.left) + countNodes(curr.right);
4  }
5
6  int[] BtToArray(BinaryTree bt) {
7      int[] array = new int[countNodes(bt.root)];
8      Queue<Pair<Node, Integer>> queue = empty;
9      if (bt.root != null) {
10         enqueue(queue, (bt.root, 0));
11         while queue is not empty {
12             (node, index) = dequeue(queue)
13             array[index] = node.data;
14             if node.left != null
15                 enqueue(queue, (node.left, 2 * index + 1))
16             if node.right != null
17                 enqueue(queue, (node.right, 2 * index + 2))
18         }
19     }
20     return array;
21 }
```

Precondition:

1. Every node.data is greater or equal to each child.data
 2. Binary tree is complete tree(except the last level)
2. [5p] Write an algorithm that converts an array into a binary tree. What are the preconditions over the array such that resulting binary tree has the max-heap property?

```
1  BinaryTree arrayToBt(int[] array) {
2      if(array == null || array.length == 0) return new BinaryTree(null);
3      Node[] nodes = new Node[array.length];
4      for (int i = 0; i < array.length; i++)
5          nodes[i] = new Node(array[i]);
6      for(int i = 0; i < array.length; i++) {
7          int left = 2 * i + 1;
8          int right = 2 * i + 2;
9          if (left < array.length) nodes[i].left = nodes[left];
10         if (right < array.length) nodes[i].right = nodes[right];
11     }
```

```
11     }
12     BinaryTree bt = new BinaryTree();
13     bt.root = nodes[0];
14     return bt;
15 }
```

Precondition:

1. For all i , $\text{array}[i] \geq \text{array}[2i+1]$ and $\text{array}[i] \geq \text{array}[2i+2]$ (when those children exist)
2. Array represents a complete binary tree